Answers to the owner's questions (kept so they don't need asking again)

Updated 2026-11-13. Newest first within each section. The PDF next to this file (docs/Answers.pdf) is made from it by Tools/make_answers_pdf.py.

The ground: puddles, cracks, tyre marks, edges, the apron

Are cracks, puddles and tyre marks part of the asphalt picture? No. The master ground material works in separate layers: (1) the surface itself (the asphalt or concrete scan, or a picture), (2) cracks and tar sealant, (3) tyre rubber, (4) wetness and puddles on top. Changing the asphalt is one setting (a new picture); every other layer stays and works on the new surface straight away. Nothing has to be redone.

If the apron is made wider or deeper later, do the details have to be drawn again? No. Cracks, sealant, wear, puddles and tyre marks are placed by world position; the runway's tyre marks by the runway's own line; the gate dirt and markings by the stand positions in the layout file. A bigger or moved apron gets everything in the right places by itself.

Is the ground seamless from runway to taxiway to apron to the stands? Yes: all of it uses the same world-space patterns, so cracks, stains and textures run straight across from one piece to the next. Where the material itself changes (asphalt to concrete), the joint looks like a real construction joint.

One master ground material: every paved surface (runway, taxiways, aprons, markings, the terminal's apron, the rental car park, and any new airport or road) is a version of one master, so every effect is made once and shows everywhere. The terminal apron keeps its own picture.

Tyre marks: strong in the runway's touchdown zones (hard landings and braking), faint along some stands' lead-in lines only, each different, much weaker than on the runway.

Gate dirt: dirt and stains at the gates (spills, engine starts), never the same at two gates.

Puddles on other airports (30,000+): they live in the ground material, so every airport built with the game's ground gets them automatically; the number of airports makes no difference to performance.

Drying in an Iceland-like climate (5-12 °C, no snow): thin wet film on pavement 1-3 h with sun and wind, 4-8 h overcast and still; puddles half a day to 2 days, shrinking from the edges; drops on windows 1-3 h (sheltered glass longer). Tunable in Settings (drying time).

Reflections: the puddles reflect the real sky now; aircraft and buildings in the water need the game's sky light fixed (a separate lighting job; it improves reflections everywhere).

Clothes

Can a garment be changed later (another shirt, other shoes)? Yes, without redoing everything: each garment piece is its own part with its own material; a new one goes through the same route in about 10 minutes, and the wardrobe picks it up.

Many colours of the same garment? Yes, without remodelling: colours and fabrics are settings on the same mesh; patterns (stripes, checks, prints) are a picture on it. Only a different cut needs new modelling.

Will the clothes fit bigger people? They are made on your body; Epic's resizing fits them to other bodies (the AI people) - tested when the crew uniforms go on the people.

Prints on the t-shirt: the print is its own high-detail layer; any image or text can be swapped in later.

How the work runs

Stopping: safe at any time; finished parts are committed as they go, so a stop loses at most the step in progress.

Releases: collected, except the puddles (0.17.10, released on its own at the owner's request). A release costs about 1-1.5 hours (packaging, testing, upload).

The MCP server (Unreal's link for Claude Code): lets changes be made and seen in the open editor in seconds instead of a hidden editor run (1-2 min) plus a test launch (3-5 min) each time: faster material tuning, placing things, laying out roads and the town. It doesn't speed up Blender modelling or packaging.

Gate areas deeper: about 1 day for the gate area alone, 1.5-2 days if the building grows towards the apron (jetways, stands and apron lines move with it), synced with the Android edition.